

Introduction to Machine Learning

What is Machine Learning?

Machine Learning (ML) is a part of **Artificial Intelligence (AI)**.

- **Artificial Intelligence (AI)** → Makes machines behave intelligently like humans.
- **Machine Learning (ML)** → A branch of AI that allows machines to **learn from data** and improve automatically.

In simple words:

Instead of programming every rule manually, we give the machine **data**, and it learns patterns from it.

How Machine Learning Works

Traditional Programming:

- We write rules.
- Computer follows rules.
- Output is generated.

Machine Learning:

- We give **data + answers**.
- Machine finds patterns.
- Machine makes predictions on new data.

Simple Example

Suppose you want to build a system that identifies spam emails.

- Traditional way: Write hundreds of rules (if email contains “win money” → spam).
- ML way: Give the system thousands of spam and non-spam emails.
- The system learns patterns and identifies spam automatically.

Key Idea of Machine Learning

ML enables systems to learn from data without being explicitly programmed.

This means:

- No need to hard-code every rule.
- System improves as more data is provided.
- Performance increases over time.

Common Applications of Machine Learning

Machine Learning is used everywhere today. Some common applications are:

Image Recognition

- Identifying objects in images.
- Face recognition in smartphones.
- Medical image analysis (detecting diseases).

Example:

- Facebook automatically tagging people in photos.
- Self-driving cars detecting traffic signs.

Natural Language Processing (NLP)

NLP helps machines understand and process human language.

Examples:

- Chatbots (like ChatGPT)
- Google Translate
- Voice assistants like Siri and Alexa

Other Real-Life Examples

- Netflix recommending movies
- Amazon recommending products
- Fraud detection in banks
- Self-driving cars

Types of Machine Learning

Machine Learning algorithms are mainly divided into **3 categories**:

Supervised Learning

The model learns from **labeled data**.

Labeled data means:

- Input data + Correct output is already given.

How it works:

- The system learns from examples.
- Then predicts output for new data.

Example:

Input (Study Hours)	Output (Pass/Fail)
2 hours	Fail
8 hours	Pass

The model learns the pattern and predicts results for new students.

Common Uses:

- Spam detection
- House price prediction
- Disease prediction

Unsupervised Learning

The model learns from **unlabeled data**.

- No correct answers are given.
- The system finds patterns or groups on its own.

Example:

Suppose we have customer data:

- Age
- Income

- Spending habits

The algorithm groups similar customers together.

Common Uses:

- Customer segmentation
- Market research
- Data clustering

Reinforcement Learning

The model learns through **trial and error**.

- It interacts with an environment.
- Gets **reward** for correct action.
- Gets **penalty** for wrong action.
- Learns to maximize rewards.

Example:

- Training a robot to walk.
- Teaching a computer to play chess.
- Self-driving cars learning driving behaviour.

Like training a dog:

- Good action → Treat (reward)
- Bad action → No treat (penalty)

Common Applications of Machine Learning (ML)

Machine Learning (ML) is widely used in many real-world applications. It helps systems learn from data and make intelligent decisions without being explicitly programmed. Below are the most important and commonly used applications explained in simple language.

1. Image Recognition

Image recognition allows computers to identify objects, people, places, or patterns in images.

How it works:

- The system is trained using thousands or millions of labeled images.
- It learns patterns like shapes, colors, edges, and textures.
- It can then recognize similar objects in new images.

Examples:

- Face unlock in smartphones
- Detecting tumors in medical scans
- Self-driving cars identifying traffic signs
- Security systems detecting intruders

2. Natural Language Processing (NLP)

Natural Language Processing helps machines understand and process human language (text or speech).

What it can do:

- Understand meaning of sentences
- Translate languages
- Answer questions
- Detect emotions in text

Examples:

- Voice assistants like **Amazon Alexa**
- Chatbots like **ChatGPT**
- Language translation tools like **Google Translate**
- Email spam detection

3. Recommendation Systems

Recommendation systems suggest products, movies, music, or content based on user behavior.

How it works:

- ML analyzes your past activity.
- It finds patterns in your preferences.
- It suggests similar items.

Examples:

- Movie recommendations on **Netflix**
- Product suggestions on **Amazon**
- Video suggestions on **YouTube**

4. Healthcare and Medical Diagnosis

ML helps doctors in diagnosing diseases and predicting health risks.

Applications:

- Detecting cancer from X-rays or MRI scans
- Predicting heart disease risk
- Drug discovery
- Personalized treatment plans

Example:

An ML model can analyze thousands of patient records and predict whether a patient is at risk of diabetes.

5. Fraud Detection

Banks and financial institutions use ML to detect suspicious transactions.

How it works:

- The system learns normal spending behavior.
- If unusual activity occurs, it raises an alert.

Examples:

- Detecting credit card fraud
- Identifying fake transactions
- Monitoring suspicious banking activity

6. Self-Driving Cars

ML is a core technology behind autonomous vehicles.

What ML does in self-driving cars:

- Detects pedestrians and vehicles
- Recognizes traffic lights and signs
- Makes driving decisions

Example:

Cars developed by **Tesla** use ML to improve autonomous driving features.

7. Speech Recognition

Speech recognition converts spoken language into text.

Examples:

- Voice typing in smartphones
- Virtual assistants
- Customer support automation

The system learns from thousands of hours of recorded speech to improve accuracy.

8. Predictive Analytics

Predictive analytics uses historical data to predict future outcomes.

Applications:

- Stock market prediction
- Sales forecasting
- Weather forecasting
- Demand prediction in businesses

Example:

Retail companies use ML to predict how much product will be sold next month.

9. Customer Segmentation

Businesses use ML to group customers based on behavior and preferences.

Why it is useful:

- Helps in targeted marketing
- Improves customer satisfaction

- Increases sales

Example:

An online store may group customers into:

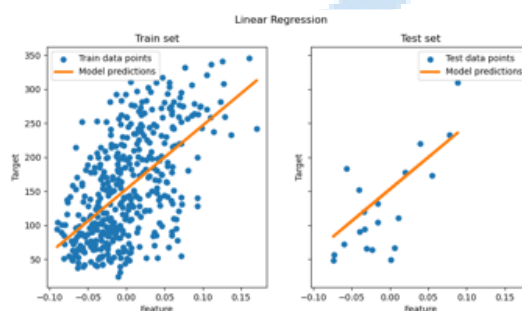
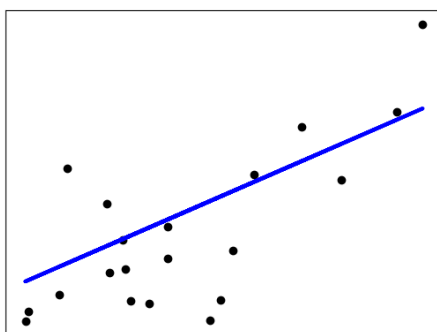
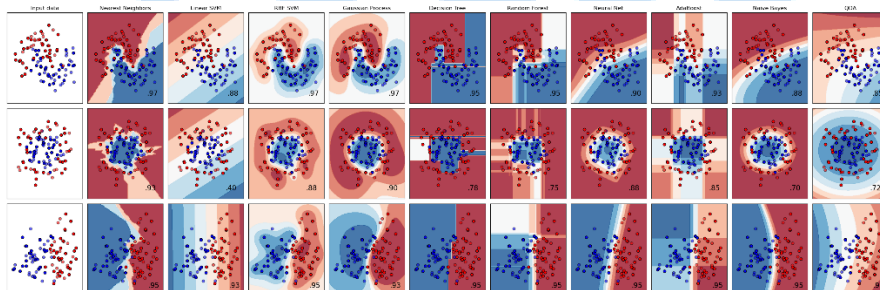
- Budget buyers
- Premium buyers
- Frequent shoppers

Libraries Used for ML

In Machine Learning and Data Science, libraries make work easier by providing ready-to-use tools and algorithms. Instead of writing everything from scratch, we use powerful libraries to build, train, and evaluate models efficiently.

Below are the most important libraries you should know, explained in simple language.

1. Scikit-Learn



Scikit-learn is one of the most popular ML libraries for beginners.

Why it is important:

- Very easy to use
- Beginner-friendly
- Great documentation
- Built on top of NumPy and SciPy

What it is used for:

- Regression (predicting continuous values like house prices)
- Classification (spam detection, disease prediction)
- Clustering (grouping customers)
- Data preprocessing
- Model evaluation

Example:

If you want to predict house prices:

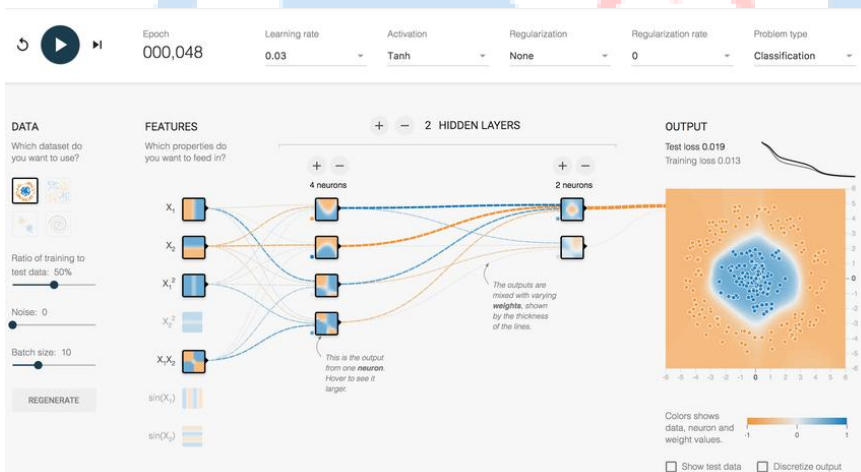
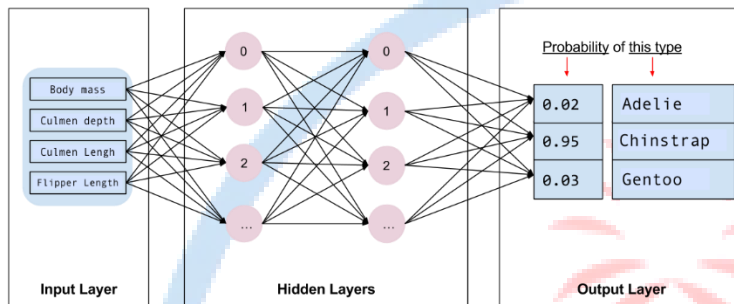
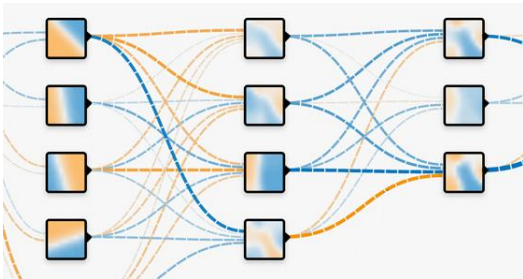
- Load data
- Choose a regression model
- Train model using Scikit-learn
- Evaluate accuracy

It provides ready-to-use algorithms like:

- Linear Regression
- Logistic Regression
- Decision Trees
- K-Means
- Support Vector Machines

Best for: Traditional Machine Learning problems.

2. TensorFlow (For Prod Ready Deployments)



TensorFlow is a powerful ML framework developed by **Google**.

Why it is important:

- Used for large-scale Machine Learning
- Good for Deep Learning
- Can run on CPUs, GPUs, and TPUs
- Production-ready

What it is used for:

- Neural networks

- Image recognition
- Speech recognition
- Large AI systems

Example:

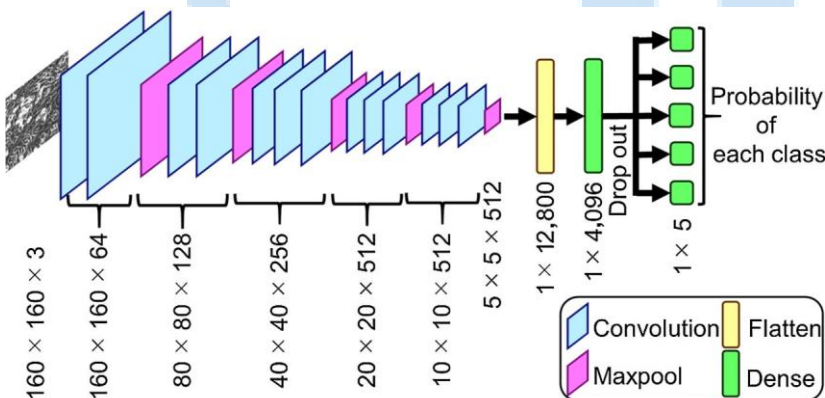
Companies use TensorFlow to train complex deep learning models for image detection or language translation.

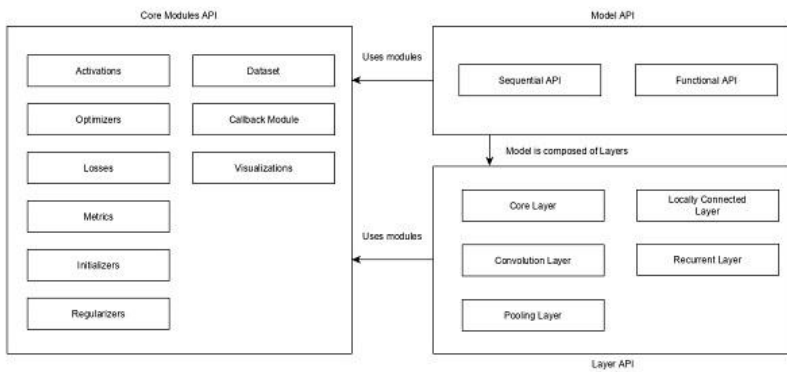
Best for: Large-scale and production-level deep learning systems.

3. Keras (For Beginner Level DL)



```
14 model.add(Dense(8, activation='relu', input_dim=4))
15 model.add(Dense(16, activation='relu'))
16 model.add(Dense(32, activation='relu'))
17 model.add(Dense(64, activation='relu'))
18 model.add(Dense(32, activation='relu'))
19 model.add(Dense(16, activation='relu'))
20
21 # 0 = low
22 # 1 = high
23
24 model.add(Dense(1, activation='sigmoid'))
25
26 model.compile(
27     optimizer='adam',
28     loss='binary_crossentropy',
29     metrics=['accuracy'])
30
31
32 x_train = np.array([
33     [1, 2, 3, 4],
34     [4, 6, 1, 2],
35     [10, 9, 10, 11],
36     [101, 95, 89, 111],
37     [90, 100, 101, 102],
38     [105, 111, 109, 102]
39 ])
40
41 # 0 = low
42 # 1 = high
43
- loss: 0.0029 - acc: 1.0000 - val_loss: 0.0041 - v
al_acc: 1.0000
Epoch 95/100
6/6 [=====] - 0s 1ms/step -
loss: 0.0028 - acc: 1.0000 - val_loss: 0.0037 - val
_acc: 1.0000
Epoch 96/100
6/6 [=====] - 0s 1ms/step -
loss: 0.0026 - acc: 1.0000 - val_loss: 0.0035 - val
_acc: 1.0000
Epoch 97/100
6/6 [=====] - 0s 813us/step -
loss: 0.0025 - acc: 1.0000 - val_loss: 0.0033 - v
al_acc: 1.0000
Epoch 98/100
6/6 [=====] - 0s 849us/step -
loss: 0.0023 - acc: 1.0000 - val_loss: 0.0032 - v
al_acc: 1.0000
Epoch 99/100
6/6 [=====] - 0s 884us/step -
loss: 0.0022 - acc: 1.0000 - val_loss: 0.0030 - v
al_acc: 1.0000
Epoch 100/100
6/6 [=====] - 0s 799us/step -
loss: 0.0021 - acc: 1.0000 - val_loss: 0.0028 - v
al_acc: 1.0000
keras_neural_network
```





Keras is a high-level API for deep learning.

Important:

- It comes built-in with TensorFlow.
- It makes deep learning much easier.

Why beginners like Keras:

- Simple and clean syntax
- Less code required
- Easy to build neural networks

Example:

To build a neural network in Keras:

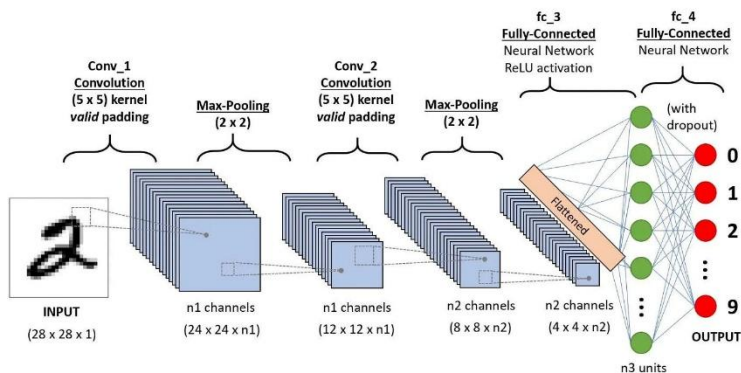
- Define model
- Add layers
- Compile model
- Train model

All of this can be done in just a few lines of code.

Best for: Beginners starting with Deep Learning.

If you are just starting deep learning, Keras is your best choice.

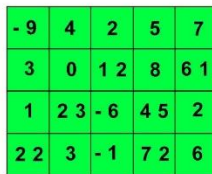
4. PyTorch (Similar to TensorFlow)



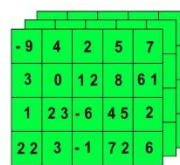
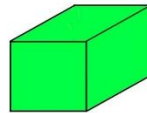
1D TENSOR /
VECTOR



2D TENSOR /
MATRIX



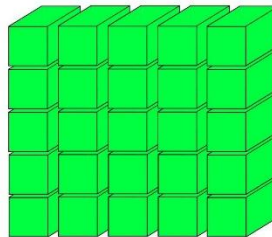
3D TENSOR /
CUBE



4D TENSOR
VECTOR OF CUBES



5D TENSOR
MATRIX OF CUBES



PyTorch is a deep learning framework developed by **Facebook** (now Meta).

Why it is popular:

- Very flexible
- Easier debugging
- Popular in research community
- Dynamic computation graph

Where it is used:

- Research projects
- Advanced deep learning
- Computer vision
- NLP models

Many AI researchers prefer PyTorch because it feels more like normal Python code.

Best for: Research and advanced deep learning projects.

Machine Learning (ML) vs Deep Learning (DL)

Machine Learning and Deep Learning are closely related, but they are not the same.

Deep Learning is actually a subset of Machine Learning.

In simple terms:

- **Machine Learning (ML)** includes techniques like classification, regression, and clustering.
- **Deep Learning (DL)** is a special type of ML that uses neural networks with many layers.

What is Machine Learning?

Machine Learning is a field of AI where machines learn patterns from data and make predictions.

Main Types of ML Problems

1. Classification
 - Predicts categories or labels
 - Example: Spam or Not Spam
2. Regression
 - Predicts continuous values
 - Example: Predicting house prices

3. Clustering

- Groups similar data points
- Example: Customer segmentation

Example

If you want to predict whether a student will pass or fail:

- Input: Study hours
- Output: Pass/Fail
- Algorithm: Logistic Regression (ML)

ML algorithms often require:

- Manual feature selection
- Human involvement in data preparation

Common ML algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- K-Means
- Support Vector Machines

What is Deep Learning?

Deep Learning is a subset of Machine Learning that uses Artificial Neural Networks.

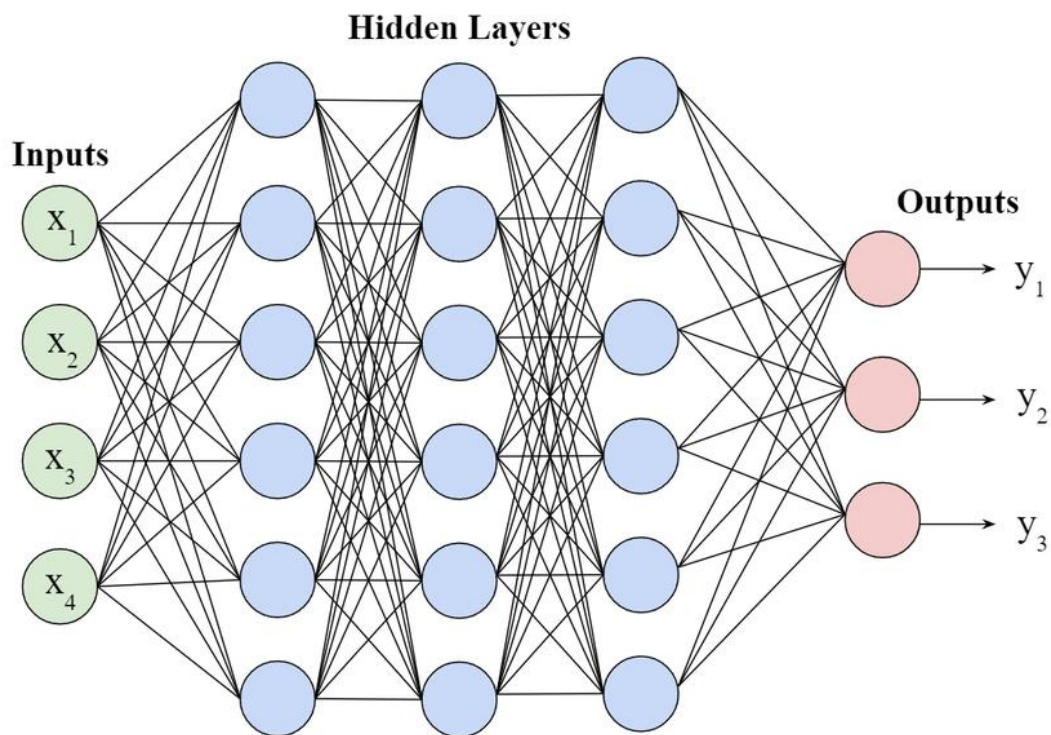
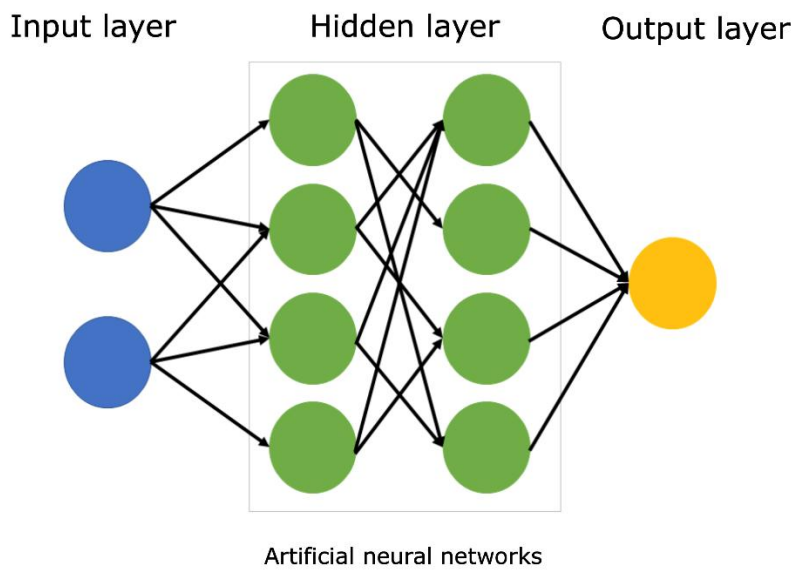
It is inspired by the structure of the human brain.

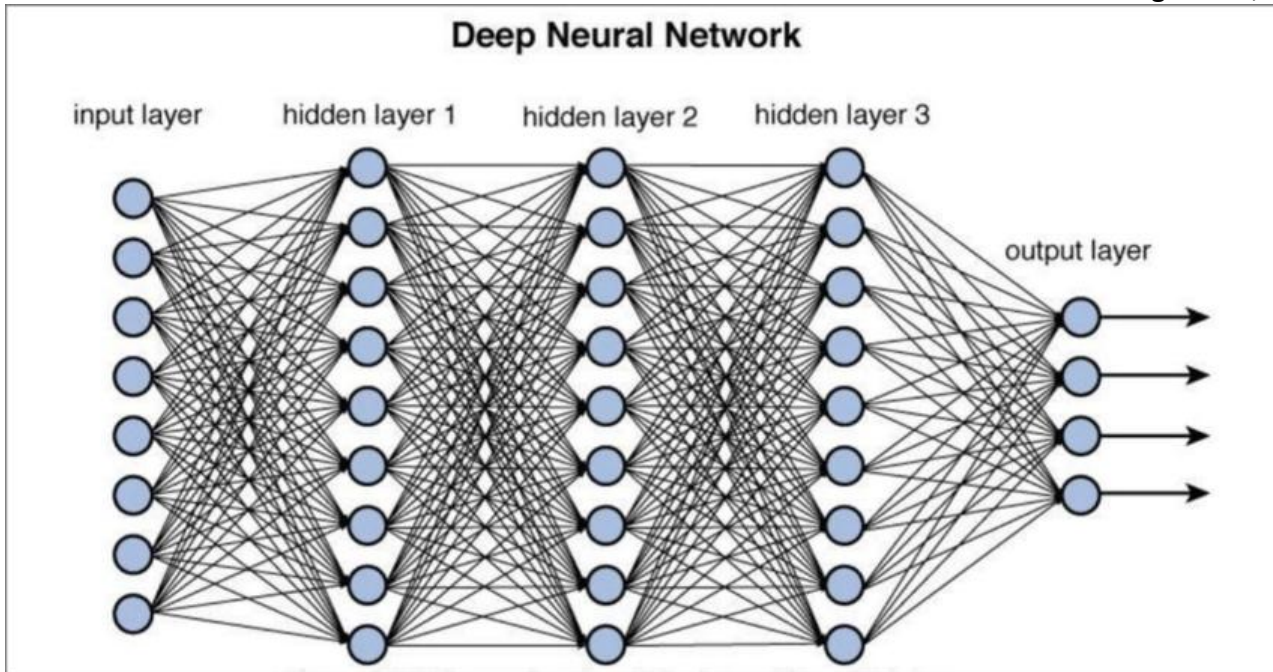
Key Idea

Deep Learning = Neural Networks with many hidden layers.

The word "Deep" means multiple layers in the network.

Neural Network Structure





A basic neural network has:

- Input Layer
- Hidden Layers
- Output Layer

Each layer contains neurons that process information.

Example

Image Recognition:

- ML approach:
 - Manually extract features like edges, shapes, colors.
 - Then apply classification algorithm.
- DL approach:
 - Feed raw image pixels.
 - Neural network automatically learns features.
 - Produces output (e.g., Cat or Dog).

Deep Learning reduces the need for manual feature engineering.

When to Use ML vs DL

Use Machine Learning when:

- Dataset is small
- Problem is simple
- Data is structured (tables, spreadsheets)
- Fast training is needed

Use Deep Learning when:

- Large dataset is available
- Working with images, speech, or text
- High accuracy is required
- Complex pattern recognition is needed

5. Relationship Between ML and DL

Artificial Intelligence

→ Machine Learning

→ Deep Learning

So:

Deep Learning is inside Machine Learning.

Machine Learning is inside Artificial Intelligence.

AI vs ML vs DL

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
Meaning	Machines that act smart like humans	Machines that learn from data	Machines that learn using neural networks
Relationship	Parent field	Subset of AI	Subset of ML
How it Works	Can use rules or learning	Learns patterns from data	Uses multi-layer neural networks

Main Idea	Make machines intelligent	Learn from data	Learn automatically using deep neural networks
Data Requirement	Not always required	Requires data	Requires large amount of data
Example Technique	Rule-based systems	Classification, Regression	Neural Networks

Example:

Technology	Role in Self-Driving Car
AI	Makes the car behave intelligently (decide when to stop or turn)
ML	Learns from driving data to improve decisions
DL	Uses neural networks to recognize traffic signs, pedestrians, and vehicles from images

In More Simple Terms:

Artificial Intelligence (AI) means making machines smart so they can think and act a little like humans. For example, a self-driving car deciding when to stop at a red light is AI.

Machine Learning (ML) is a part of AI where machines learn from data instead of being told every rule. For example, Netflix suggesting movies based on what you watched before is ML.

Deep Learning (DL) is a part of ML that uses something called neural networks (inspired by the human brain) to learn from a lot of data. For example, face unlock in your phone recognizing your face is Deep Learning.

The Need of Machine Learning

One of the earliest real-world uses of Machine Learning was email spam filtering.

In the '90s, email inboxes were full of unwanted messages like:

- Fake lottery wins
- Miracle diet plans
- Suspicious job offers

Instead of writing fixed rules like:

- “If email contains the word ‘lottery’, mark it as spam”

Engineers tried something smarter.

They:

- Collected many spam emails
- Collected many normal emails (called “ham”)
- Let the system learn the difference from examples

Over time:

- Users kept marking emails as spam.
- The system improved.
- It started detecting new types of spam automatically.

That was Machine Learning in action.

What Machine Learning Is NOT

If you download 500 books onto your computer, does it become smart?

No.

It just stores the data.

It cannot:

- Write like Shakespeare
- Summarize the books
- Understand the meaning

That is just data storage.

Machine Learning is not about storing data.
It is about learning from data.

What Machine Learning IS

Machine Learning is the science of teaching computers to learn from data and improve over time without being explicitly programmed for every situation.

Arthur Samuel defined it as:

“Giving computers the ability to learn without being explicitly programmed.”

Tom Mitchell explained it in a more detailed way:

A program learns from:

- Experience (E)
- At a Task (T)
- Measured by Performance (P)

If performance improves with experience, it is learning.

Breaking It Down with the Spam Example

Task (T):

Identify whether an email is spam or not.

Experience (E):

A collection of emails labeled as spam or not spam.

Performance (P):

Accuracy — how many emails are correctly classified.

If the system becomes better at detecting spam after seeing more emails, it is learning.

Important Terms in Machine Learning

Training Set:

The collection of examples used to teach the system.

Training Instance (Sample):

One single example in the dataset.

Model:

The part of the system that learns patterns and makes predictions.
Think of the model as the brain of the system.

Some common models include:

- Neural Networks
- Random Forests
- Decision Tree
- Naive Bayes classifier
- Support Vector Machines

Traditional Programming vs Machine Learning**Traditional Programming**

In traditional programming:

Human writes all the rules.

Formula:

Input + Rules → Output

Example:

If email contains “free money”, mark as spam.

Problem:

What if spammers write:

- “frëe m0ney”
- “freee money”
- “get money fast”

The rule breaks easily.

Machine Learning Approach

In Machine Learning:

We give:

Input + Correct Output (examples)

The system learns the rules by itself.

Process:

Input + Output examples → Learning → Model

New Input + Model → Prediction

Instead of hardcoding every rule, the system:

- Learns patterns
- Understands similarities
- Adapts to new variations

That is why ML systems improve over time.

Why Machine Learning Is Powerful

- It adapts as new data comes.
- It improves with experience.
- It handles complex patterns better than manual rules.
- It works even when rules are too difficult to write.